Truth-conditional Semantics and the Fregean Program

Martina Barili / 12 Oct.

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1 Truth-conditional Semantics

In this semantic theory, knowing the meaning of a sentence is to know what the world would have to be like for the sentence to be true.

Sentence	**Truth-conditions**
"This is a Reading Group on Semantics" =	The state of affairs in the real world are conditions that make the sentence true: we are in a Reading Group on Semantics in this moment.

The above theory merely pairs sentences with truth-conditions, but does not say anything about how they come about: every meaningful part of a sentence contributes to its truth-conditions in a systematic way.

1.1 Compositionality

Frege is the first (second only to Aristotle) to interest himself on semantic composition. On his view, this happens through saturation: unsaturated functions are applyied to their arguments, so the second saturate the former. Functions are sets.

Although Frege is the first speaking about compositionality of meaning, the first workedout versions were by Lewis, Montague and Cresswell for fragments of English.

1.2 Sets and functions

1.2.1 Sets

Overview of some concepts:

- memebrs, elements $\in$;
- unique empty set $\varnothing$;
- intersection A$\cap$B;
- disjoint sets (no members in common);
- subsets A$\subset$B;
- union A$\cup$B;
- complement B-A;
- abstraction notation for sets by defining the conditions to be satisfied for an element to be in a set.

1.2.2 Functions

Functions are sets that define a relation.
A 2-place relation (function) is a set of ordered pairs. The requirement is that the second member of the pair is *uniquely deteremined* by the first.

Definition:
A relation is a function iff it satisfies the following condition: For any x: if there are y and z such that $< x, y > \in f$ and $< x, z > \in f$, then $y = z$.

- Domain
- Range
- *A onto B* when B is the range.
- *A into (or to) C* when C is a superset of the range.
- In $f(x) = y$, *y* is the *value* of *f*.

2 Executing the Fregean Program

Starting from Frege and some basic analysis of English, more complicated cases will be analysed, in an attempt to develop a semantic theory of natural language. However, compositionality and function application (Frege's Conjecture) will remain a fil rouge.
This semantic theory takes sentences to be phrase structure trees made of subparts.